Hybrid Variational Quantum Circuit with Transformer for Classifying Medical Conditions

Waseem Ahmad[1] , Kalpesh Prajapati[2] , Kameshwar Rao JV[3]
AI CoE
HCL Technologies Ltd.
Noida, India
[1]waseem.ahmad809@gmail.com, [2]kalpesh.jp89@gmail.com, [3]kameshjvkr@gmail.com

2023 IEEE International Conference on Quantum Computing and Engineering (QCE) | 979-8-3503-4323-6/23/$31.00 ©2023 IEEE | DOI: 10.1109/QCE57702.2023.10221

***Abstract*—Reviews in the form of text available on different web-based platform, can provide very important insights analyzing the consumers behavior and their wants and needs. Such data can be used to improve the quality or assess the effectiveness of pharmaceutical product. This effort aims to assess the medical conditions of a patient by analyzing the text reviews of pharmaceutical products, and thereby increase the patient's healthcare quality. Recently new computing paradigms like quantum machine learning gained a lot of attention. However, it is not yet feasible to perform quantum operations on high dimensional data using current quantum hardware. Hence a hybrid approach using both classical and quantum machine learning methods has been proposed. We compared our results with the performance of the Bidirectional Encoder Representations from Transformer (BERT) model. The result shows the promise of quantum computing technology when used in combination with a classical machine learning approach. In comparison to the current state-of-the-art technology, where the transformer-based techniques achieved 85.7% of accuracy, our hybrid approach achieved 87.1% accuracy.**

I. Poster Relevance

This poster presents the medical conditions classification using hybrid (classical-quantum) approach. It helps in detecting the diseases from the text reviews of pharmaceutical products for diagnostic decision support using the quantum machine learning with hybrid quantum variational circuits.

II. Poster : Hybrid Variational Quantum Circuit with Transformer for Classifying Medical Conditions

The main contributions of the work can be summarized as follows:

- We adopted a hybrid approach using a VQC (variational quantum circuit) combined with a traditional state-of-the-art transformer model called BERT[1] to address the text data classification task.
- The performance of the proposed method is compared with the baseline model (i.e., BERT) to verify the quantum computing efficiency. The results show that the proposed hybrid model achieved better performance compared to baseline model.

Quantum machine learning algorithms have been proposed in different forms in recent years [2]. In addition to quantum core methods, Variational Quantum Algorithms (VQA) seem to be most important in the current NISQ era for several reasons [3]. VQAs usually consist of several components, but the main part is the design of the circuit used, also called as ansatz. Furthermore, the VQA ansatz is essentially parameterized so that it can be used as a predictive model by optimizing the parameters to achieve a specific goal, i.e., to minimize a specific loss. Using techniques such as gradient descent, this model can be trained to predict the target from unseen data [4].

A. Proposed Method

This section provides a brief overview of the proposed method, which is a hybrid model that combines a conventional transformer and a variational quantum circuit. The proposed method is used for identifying a medical condition based on a patient's review about drugs. The variational quantum circuit is a promising new approach for machine learning, and it can be used to improve the performance of the transformer by taking advantage of the properties of quantum computing, such as superposition and entanglement.

The basic block diagram of the proposed hybrid architecture is shown in Fig. 1. The output dimension of the BERT model is 768. In this experiment, the data consists of 10 different diseases, so these 768 features must be mapped to 10 features that represent the probability of the corresponding disease. A fully connected layer is used to map 768 features to 10 outputs to predict the disease class. However, as shown in Fig. 1, the baseline model has been extended and fine-tuned with a variational quantum circuit. The variational quantum circuit takes the 10-dimensional data from the previous 10-dimensional dense layer and outputs the same 10 dimensions. Fig. 2 shows the 10-qubit variational quantum circuit used in the proposed work.

In this work, we initialized 10-qubit quantum circuit with $|0\rangle$ state and then applied Hadamard gate on all the qubits to make them in superposition, then angle encoding using rotation-Y gates are applied to encode the classical data into quantum state in the quantum data encoding layer. Here, the parameter (i.e., rotation value) of the rotation-Y gate is set with a classical output data obtain from the fully connected layer of previous stage. After encoding the classical data into a quantum state, a parameterized quantum circuit is used to map the encoded qubit state to the final state representing the probability of the corresponding disease. As illustrated in the Fig. 2, entangle layer (i.e., represented as Entanglement CNOT) and parameterized layer (i.e., represented as Rotation Y and Rotation X) are used five times in the parameterized quantum circuit (i.e., CNOT > $R_y(\theta)$ > CNOT > $R_x(\theta)$ repeated 5 times). In a parameterized quantum layer, rotation gates are used whose parameters are trained during training through back-propagation. Additionally, controlled NOT

979-8-3503-4323-6/23/$31.00 ©2023 IEEE
DOI 10.1109/QCE57702.2023.10221

Authorized licensed use limited to: Universita degli Studi di Salerno. Downloaded on July 02,2026 at 10:14:32 UTC from IEEE Xplore. Restrictions apply.

gates are used in entangle layer which entangles the state of the two neighboring qubits alternatively (i.e., CNOT between qubit pairs 0-1, 2-3, 4-5, 6-7, 8-9). Last, the quantum state of each qubit is measured with respect to Pauli Z-axis to get the likelihood values for all 10 diseases from the respective 10-qubit measurements.

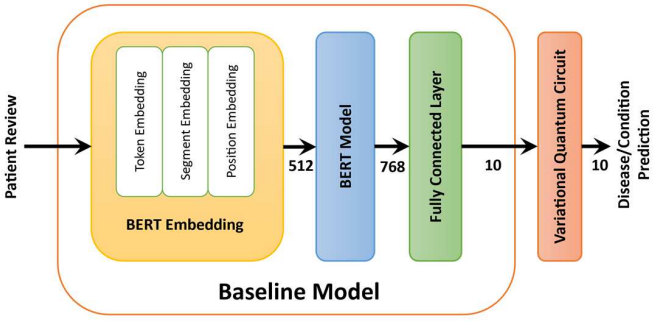


Fig. 1: Diagram of the proposed model

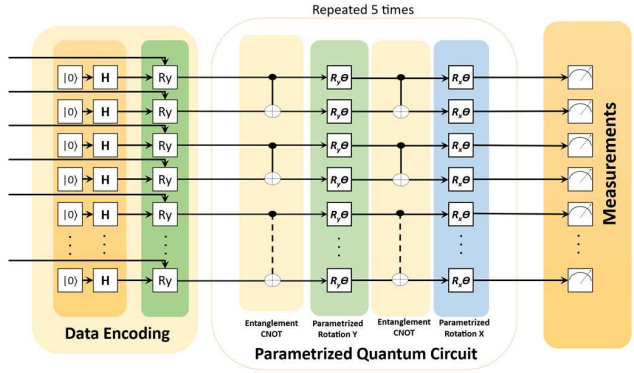


Fig. 2: The variational quantum circuit used in the proposed method

B. Experimental Analysis

Dataset Description

This work uses a subset of the UCI ML Drug Review dataset [5]. The original dataset consists of 916 unique medical conditions but only 10 medical conditions are considered in this work. Additionally, to make the problem easier, only 500 reviews are taken from each medical condition which results in to total 5000 samples. Out of which, 80% of samples are used for the training while rest 20% samples are used for testing purpose.

Implementation Details

The proposed technique has been implemented on PyTorch framework with the help of the TorchQuantum library [6] . The models have been trained up to 100 epochs with a batch size of 8. The Adam optimizer has been used with learning rate of 2×10^{-6}. Furthermore, the parameter shift-rule [7] has been used to find the gradients of the parameters used in variational quantum circuit. Here we used cross-entropy loss to train the model.

Evaluation

This section presents a comparison of the proposed hybrid method with the baseline classical machine learning method. As we discussed earlier BERT model has been used as a baseline model. Here, the performance is compared in terms of accuracy, precision, recall and F1-score and same is presented in Table 1.

TABLE 1: QUANTITATIVE COMPARISON OF PRSSSOPOSED METHOD WITH BASELINE

Method	Accuracy	Precision	Recall	F1_Score
BERT	85.7%	0.8593	0.8570	0.8562
Proposed Hybrid QBERT Model	**87.1%**	**0.8733**	**0.8710**	**0.8708**

Fig. 3 describes the confusion matrix on the test data for the proposed model. Here, it is observed that Acne, Anxiety, birth control and depression are the classes which are highly conflicted with birth control, depression, acne and bipolar disorder respectively with more than 8 false predictions in sample of 100.

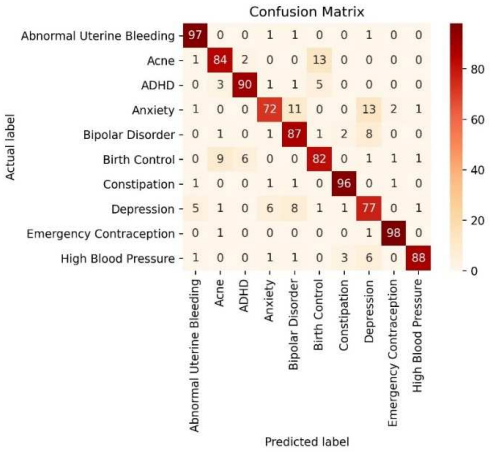


Fig. 3: Confusion matrix of the proposed hybrid BERT model

C. Conclusions

Medical data contains a large amount of information that can be used to improve patient's technological experiences through machine learning techniques. The energy and cost required to analyze such a large amount of data motivates us to use machine learning techniques to partially mitigate the limitation. Recently many quantum machine learning studies showed performance equal to or better than classical machine learning approaches. Here we tested the efficacy of unique properties of quantum computing with traditional ML which improves the overall performance. Considering these factors, a hybrid strategy combining deep learning and variational quantum circuits has been used in this study. For all quantitative metrics such as precision, accuracy, recall and F1 score, the proposed hybrid technique outperformed the baseline.

REFERENCES

[1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," ArXiv, vol. abs/1810.04805, 2019.

[2] Biamonte, J., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., and Lloyd, S. (2017). Quantum machine learning. Nature, 549(7671):195–202

[3] Cerezo, M., Arrasmith, A., Babbush, R., Benjamin, S. C., Endo, S., Fujii, K., McClean, J. R., Mitarai, K., Yuan, X., Cincio, L., et al. (2021). Variational quantum algorithms. Nature Reviews Physics, 3(9):625–644.

[4] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii, "Quantum circuit learning," Physical Review A, 2018.

[5] F. Graßer, S. Kallumadi, H. Malberg, and S. Zaunseder, "Aspect-based ¨ sentiment analysis of drug reviews applying cross-domain and cross-data learning," Proceedings of the 2018 International Conference on Digital Health, 2018.

[6] H. Wang, Y. Ding, J. Gu, Y. Lin, D. Z. Pan, F. T. Chong, and S. Han, "Quantumnas: Noise-adaptive search for robust quantum circuits," ArXiv, vol. abs/2107.10845, 2021.

[7] J. Li, X. dong Yang, X. Peng, and C. pu Sun, "Hybrid quantum-classical approach to quantum optimal control." Physical review letters, vol. 118 15, p. 150503, 2017

Authorized licensed use limited to: Universita degli Studi di Salerno. Downloaded on July 02,2026 at 10:14:32 UTC from IEEE Xplore. Restrictions apply.